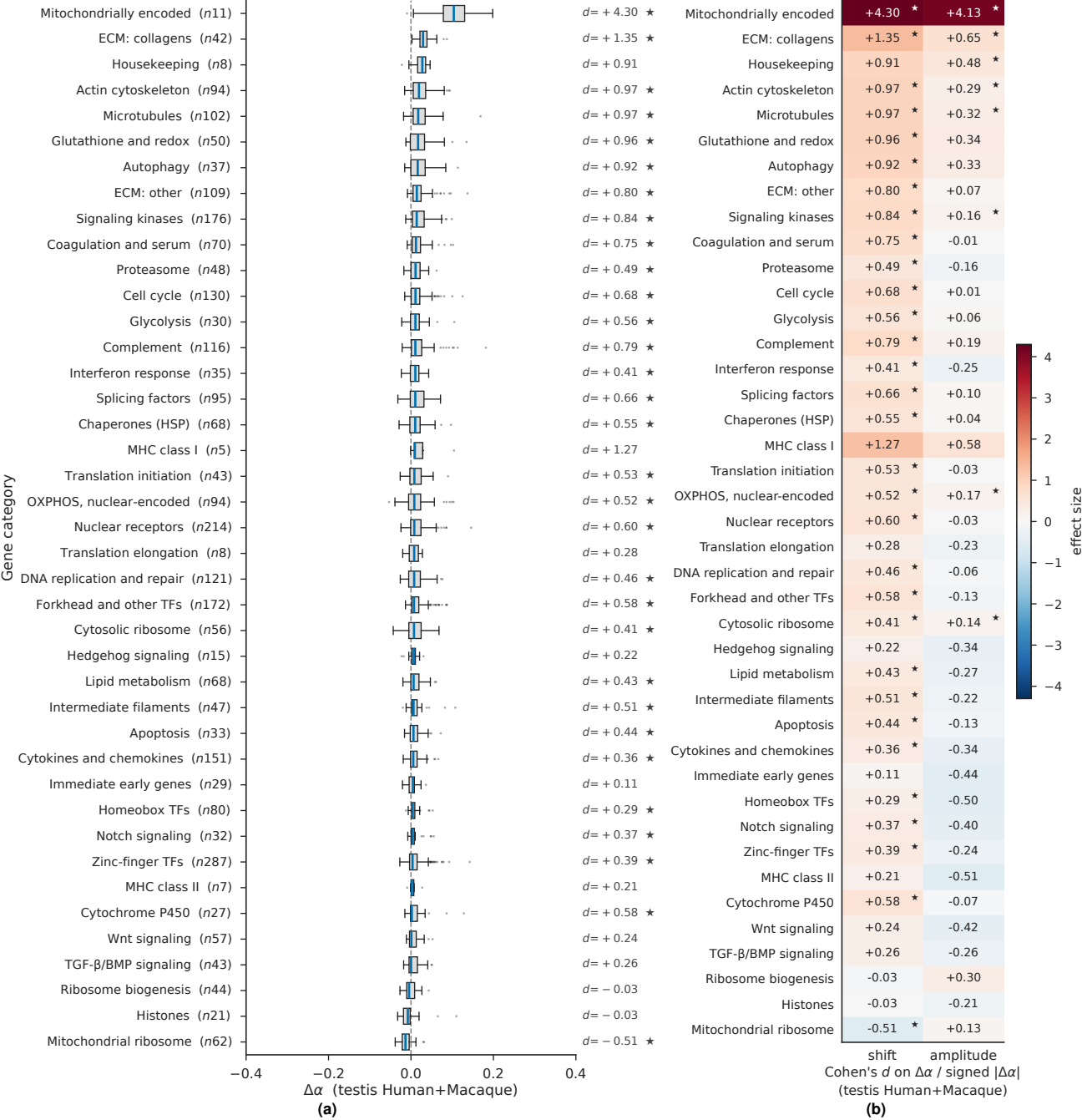

Gene category
Mitochondrially encoded (n11) d= + 4.30 ★
ECM: collagens (n42) d= + 1.35 ★
Housekeeping (n8) d= + 0.91
Actin cytoskeleton (n94) d= + 0.97 ★
Microtubules (n102) d= + 0.97 ★
Glutathione and redox (n50) d= + 0.96 ★
Autophagy (n37) d= + 0.92 ★
ECM: other (n109) d= + 0.80 ★
Signaling kinases (n176) d= + 0.84 ★
Coagulation and serum (n70) d= + 0.75 ★
Proteasome (n48) d= + 0.49 ★
Cell cycle (n130) d= + 0.68 ★
Glycolysis (n30) d= + 0.56 ★
Complement (n116) d= + 0.79 ★
Interferon response (n35) d= + 0.41 ★
Splicing factors (n95) d= + 0.66 ★
Chaperones (HSP) (n68) d= + 0.55 ★
MHC class I (n5) d= + 1.27
Translation initiation (n43) d= + 0.53 ★
OXPHOS, nuclear-encoded (n94) d= + 0.52 ★
Nuclear receptors (n214) d= + 0.60 ★
Translation elongation (n8) d= + 0.28
DNA replication and repair (n121) d= + 0.46 ★
Forkhead and other TFs (n172) d= + 0.58 ★
Cytosolic ribosome (n56) d= + 0.41 ★
Hedgehog signaling (n15) d= + 0.22
Lipid metabolism (n68) d= + 0.43 ★
Intermediate filaments (n47) d= + 0.51 ★
Apoptosis (n33) d= + 0.44 ★
Cytokines and chemokines (n151) d= + 0.36 ★
Immediate early genes (n29) d= + 0.11
Homeobox TFs (n80) d= + 0.29 ★
Notch signaling (n32) d= + 0.37 ★
Zinc-finger TFs (n287) d= + 0.39 ★
MHC class II (n7) d= + 0.21
Cytochrome P450 (n27) d= + 0.58 ★
Wnt signaling (n57) d= + 0.24
TGF-β/BMP signaling (n43) d= + 0.26
Ribosome biogenesis (n44) d= − 0.03
Histones (n21) d= − 0.03
Mitochondrial ribosome (n62) d= − 0.51 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (testis Human+Macaque)
(a)
Mitochondrially encoded +4.30 ★ +4.13 ★
ECM: collagens +1.35 ★ +0.65 ★
Housekeeping +0.91 +0.48 ★
Actin cytoskeleton +0.97 ★ +0.29 ★
Microtubules +0.97 ★ +0.32 ★
Glutathione and redox +0.96 ★ +0.34
Autophagy +0.92 ★ +0.33
ECM: other +0.80 ★ +0.07
Signaling kinases +0.84 ★ +0.16 ★
Coagulation and serum +0.75 ★ -0.01
Proteasome +0.49 ★ -0.16
Cell cycle +0.68 ★ +0.01
Glycolysis +0.56 ★ +0.06
Complement +0.79 ★ +0.19
Interferon response +0.41 ★ -0.25
Splicing factors +0.66 ★ +0.10
Chaperones (HSP) +0.55 ★ +0.04
MHC class I +1.27 +0.58
Translation initiation +0.53 ★ -0.03
OXPHOS, nuclear-encoded +0.52 ★ +0.17 ★
Nuclear receptors +0.60 ★ -0.03
Translation elongation +0.28 -0.23
DNA replication and repair +0.46 ★ -0.06
Forkhead and other TFs +0.58 ★ -0.13
Cytosolic ribosome +0.41 ★ +0.14 ★
Hedgehog signaling +0.22 -0.34
Lipid metabolism +0.43 ★ -0.27
Intermediate filaments +0.51 ★ -0.22
Apoptosis +0.44 ★ -0.13
Cytokines and chemokines +0.36 ★ -0.34
Immediate early genes +0.11 -0.44
Homeobox TFs +0.29 ★ -0.50
Notch signaling +0.37 ★ -0.40
Zinc-finger TFs +0.39 ★ -0.24
MHC class II +0.21 -0.51
Cytochrome P450 +0.58 ★ -0.07
Wnt signaling +0.24 -0.42
TGF-β/BMP signaling +0.26 -0.26
Ribosome biogenesis -0.03 +0.30
Histones -0.03 -0.21
Mitochondrial ribosome -0.51 ★ +0.13
shift
amplitude
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(b)
4
3
2
1
0
−1
−2
−3
−4
effect size